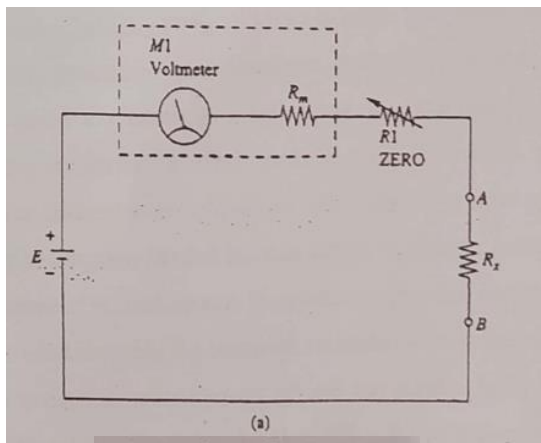


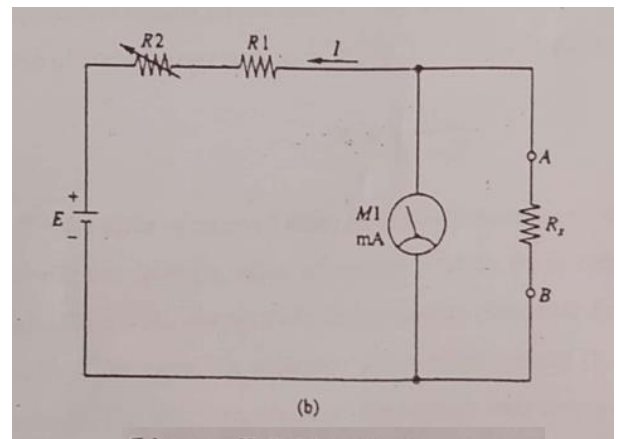
DC analog ohmmeter

A **DC analog ohmmeter measures electrical resistance** using the relationship $I = E/R$ (Ohm's law). It consists of a simple series circuit with three resistances: the voltmeter's internal resistance (R_m), a zero-calibration control resistor (R_1), and an unknown resistance (R_x).

- **The scale is reversed compared** to voltage and current scales. At maximum current, the meter reads zero.
- **Analog ohmmeters** require **manual zeroing**, while **modern digital** ones do it **automatically**.
- **Series ohmmeters** don't work well for low resistances ($< 100\Omega$); for these, parallel ohmmeters are used.
- **shunt ohmmeter circuit**, describing the use of a meter movement (milliammeter or microammeter). RI can include fixed and variable resistors and, typically $R_1 = 100 \times R_m$ or higher.



(a) series ohmmeter

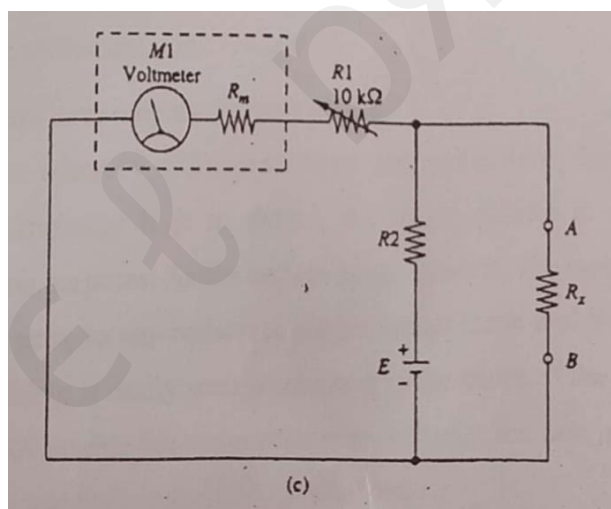


(b) parallel (short) ohmmeter,

$$R_x = I_2 R_m / (I_1 - I_2)$$

$I_1 = I$ without connecting R_x

$I_2 = I$ with connecting R_x



(c) series-parallel ohmmeter.

Ohmmeter accuracy vs. Battery aging

The accuracy of the ohmmeter **depends on the battery** How?!!

1. The output voltage from battery deflects the pointer to **the full scale (zero-calibration)**, as the battery ages the output voltage **decrease**, so the pointer can't reach to the zero ohm on the scale, so measurement will have an error.

This error can easily to notice ↓

if the pointer can't reach to the zero → change the battery.

2. The weakness of the battery causes increased internal resistance, also this problem makes the pointer not able to reach zero

The solution to any of these problems is to change the battery. Even if the meter is not used, the battery should be refreshed (changed or be charged) every 6 months or more if the manufacturer recommends.

Analog Multimeters

What's the Multimeter?

It's an instrument that has serval switch-selectable voltage, current, and resistance scales. Each switch connects the connections of his mode when it turns on, most instruments include a rectifier to allow reading AC values.

- Follow the same rules as individual meters to avoid damaging the instrument. Store the multimeter in the OFF position or set to the highest DC VOLTS range if there is no OFF position. Start with the highest range for current or voltage and work down to a readable deflection
- **passive mete (YOM)** : - the meter doesn't need a power supply to work.
- **Active meter**: - the meter needs a power supply to work.

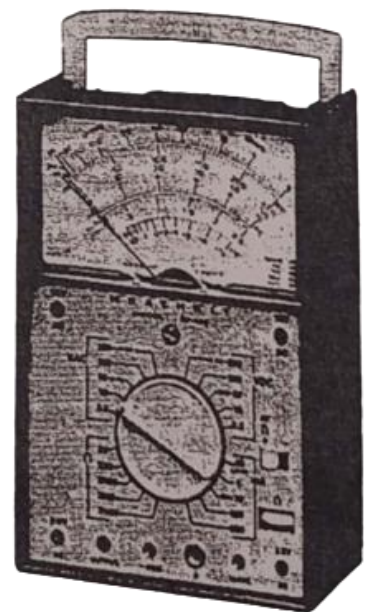


Fig. (1-12) Analog multimeter.

Analog Alternating Current (AC)

- **Basic PMMC meter movements** can't correctly indicate AC values and might be damaged by AC. Four types of meter movements that can indicate AC values are: **thermocouple, hot-wire, electrodymanometer, and iron-vane.**
- **A DC meter movement** (D'Arsonval or taut-band) can read AC values if a rectifier or RMS-to-DC converter is used between the AC input and the DC meter movement.

Which AC value is being measured?

Measuring voltage and current values in AC circuits is complex **due to their cyclical nature and different recognized values.** AC meters display specific AC values. For a **pure sine wave**, the measurable values include: **peak voltage (E_p)**, **peak-to-peak voltage (E_{p-p})**, **root mean square voltage (E_{rms})**, and **average voltage (E_{av})**.

1. **The peak voltage (E_p)** is the voltage from baseline zero to the peak of the wave.
2. **The peak-to-peak voltage (E_{p-p})** is magnitude of the voltage between the peak voltages.

$$E_{p-p} = |+E_p| + |-E_p| = 2E_p \quad \text{(if zero baseline=0v)}$$

3. **The root mean square voltage (E_{rms})** = the effective voltage. **The rms value** is the magnitude of the DC voltage that produce the same amount of heating in a resistive load.

$$E_{rms} = \sqrt{\left[\frac{1}{T} \int_0^T E(t) dt \right]^2}$$

where

E_{rms} = the root mean square value of the sine wave voltage

$E(t)$ = the instantaneous value of E with respect to time (1-14)

T = the period of the waveform, that is, the time required for one complete cycle

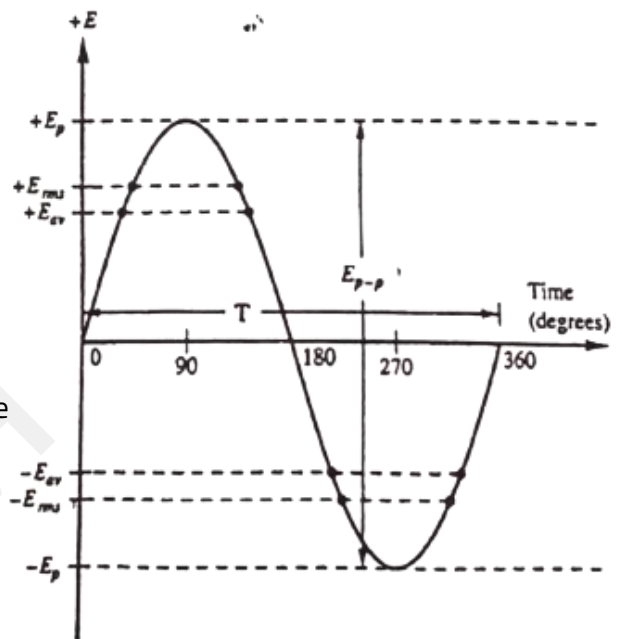


Fig. (1-13) Sine wave relationships

(if the wave is sine or cosine only) $\rightarrow E_{rms} = \frac{E_p}{\sqrt{2}} = \frac{\sqrt{2} E_p}{2} = 0.707E_p$

4. The average voltage (E_{av})

if the wave is sine or cosine only ↓

(over the entire cycle) → $E_{av} = 0$

(over the half-cycle) → $E_{av} = \frac{2E_p}{\pi} = 0.637E_p$

This is a symmetrical square wave

- $T = t_1 + t_2$
- $E_p - p = 2E_p$

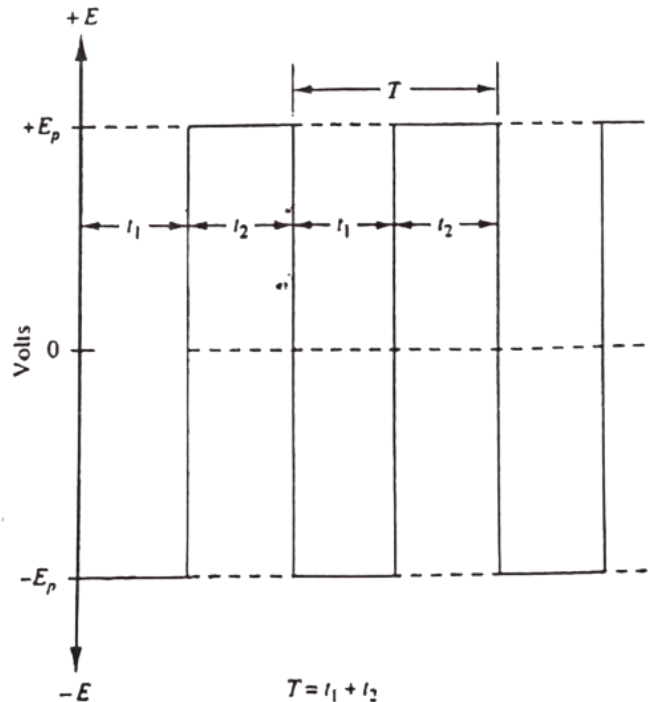


Fig. (1-14) Symmetrical square wave with baseline symmetry and left-right symmetry

Form Factor (Λ)

The form factor (Λ) of a periodic waveform is the ratio of its rms value to its half-wave average value:

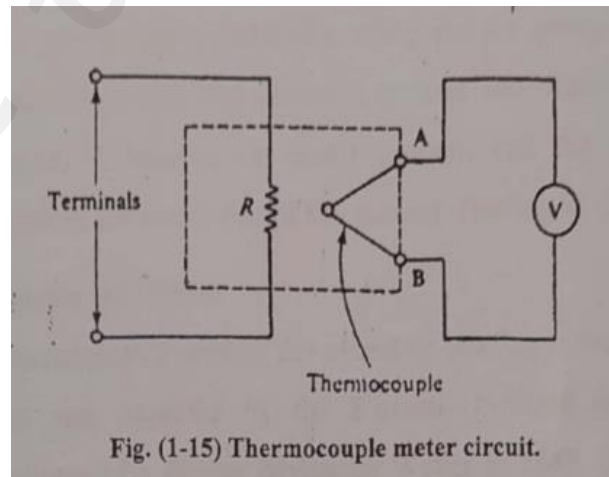
$$\Lambda = \frac{V_{rms}}{V_{av}}$$

For the case of the sine wave, the value of the form factor is

$$\Lambda = \frac{\left[\frac{\sqrt{2}E_p}{2} \right]}{\left[\frac{2E_p}{\pi} \right]} = 1.11$$

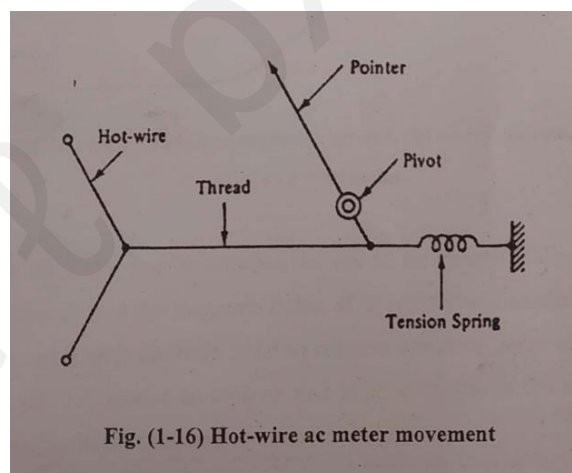
Thermocouple AC current meter

The thermocouple ammeter is used to measure RF currents. It consists of a resistive heating element and a thermocouple sensor. The ac current heats the resistive element R , increasing the thermocouple voltage. The voltmeter across terminals A and B measures this voltage, calibrated in amperes or milliamperes, reflecting the rms value. While any ac meter measures sine wave rms values, only thermocouple ammeters accurately measure non sinusoidal waveforms.



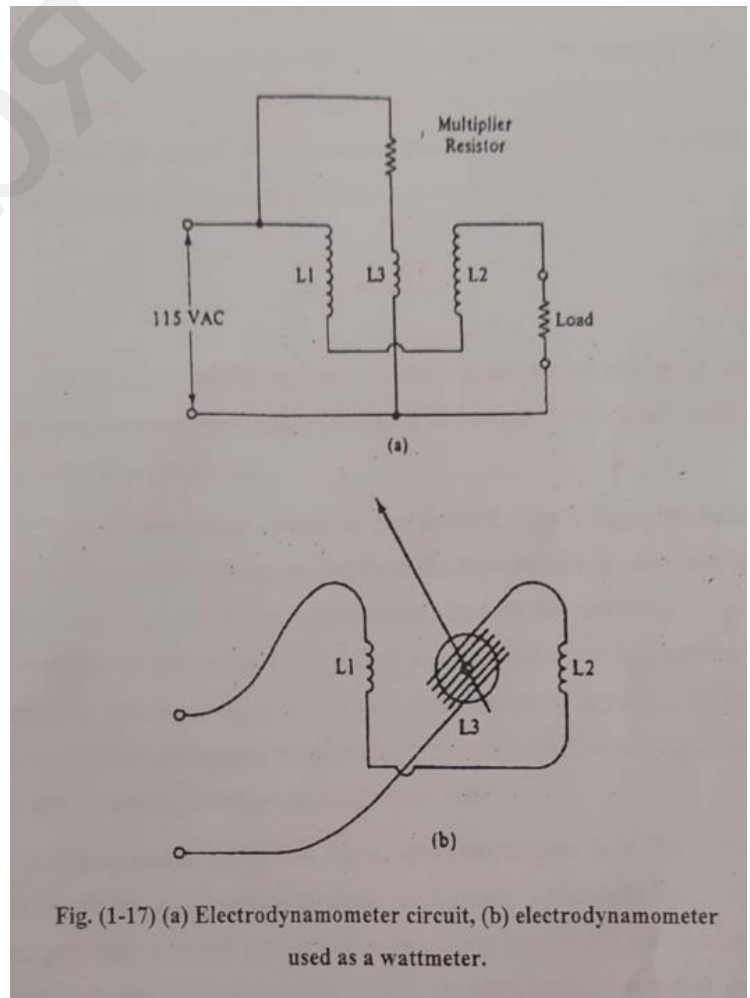
Hot-wire ammeters

The hot-wire AC meter uses a resistance wire that heats and expands when a current flows through it, causing a pointer to deflect. **This deflection is proportional to the RMS value** of the current. The meter accurately measures RMS value regardless of the waveform shape and is independent of the frequency of the AC signal.



The electrodynamicometer AC meter

- **The electrodynamicometer is one of the oldest AC meters.** This meter creates its own magnetic field from the current in the coils L1 through L3, with L1 and L2 being stationary
- **The electrodynamicometer is a square-law device,** meaning the pointer deflects in proportion to the square of the average current in its coils.
- Most electrodynamicometers are **calibrated** in terms of rms current for sine waves. They typically have a full-scale rating of 100 mA or more. $\rightarrow E_{rms} = \sqrt{i_{av}^2}$
- **Electrodynamicometer voltmeters** can be constructed from a multiplier resistor and a 100-mA meter, suitable only for low-impedance applications. (it has low sensitivity $\sim 10\Omega/V$).
- **One principal application is in wattmeters** used in 60-Hz AC power line systems.



- **The stationary coils** carry heavy current, while the movable coil, connected across the load, measures voltage.
- **When there is no multiplier resistor** & the resistance of L3 is suitable to limit the current to the same value $\rightarrow$ The deflection of the pointer connected to L3 is proportional to the product of the volt times amperes (**$V \cdot A = \text{WATT}$**)

Iron-vane meter movement

- **The iron-vane meter movement consists** of a wire coil and two iron vanes, one movable and the other stationary. The current in the coil magnetizes the vanes, causing the movable vane to deflect proportionally to the rms current.
- **It provides accurate readings** at 50 Hz (Europe) and 60 Hz (US) but faces hysteresis and eddy current losses at higher frequencies.
- **In DC, residual magnetism causes errors**, but AC eliminates this by reversing direction every half-cycle.

